Indian Institute of Technology Hyderabad
Signal Processing, Fall 2026
Quiz 5, 01.11.2026, 10 points

1. State `True` or `False`. Justify your answer.
   (a) Every *all-pass* system is a *minimum-phase* system. (2)
   (b) All *rational linear time-invariant* systems can be expressed as product of a *minimum-phase* and an *all-pass* system. (2)
   (c) Among *all* causal and stable rational linear time-invariant systems with magnitude response $|H(e^{j\omega})|$, the minimum-phase system is the one with the minimum group delay. (2)
2. When does the factorization of a rational linear time-invariant system into a minimum-phase and all-pass system help? Illustrate with an example. State any assumptions. (4)